

VALIDATION TESTING OF PRECAST TUNNEL LINING SEGMENTS USING POLYMERIC FIBERS

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ABSTRACT

Fiber reinforcement in precast tunnel segments is quite attractive since it reduces the cost and labor associated with the fabrication process while it improves the post-cracking behavior considerably. Among the benefits, one can address improvements in handling, fatigue, impact resistance and durability while reducing the crack widths significantly. Flexural tests of full-scale precast tunnel segments are conducted using a newly developed testing laboratory for large structural testing. Using the closed-form material properties obtained from the flexural beam specimens, the response of full sections is predicted and the test results are compared with the full-scale test results obtained on the testing facility. Using the proposed methodologies, one can develop proper material and structural models for the accurate design of tunnel segments due to their unique design characteristics.

KEYWORDS: Tunnel lining, Precast concrete, Fiber reinforced concrete, Polymeric Fiber, residual strength, simulation

1. INTRODUCTION

The development of excavation technology using the tunnel boring machines (TBM) has resulted in widespread use and study of precast concrete tunnel linings (PCTLs). In addition to easier handling and installation processes, higher quality and more economical production of PCTL can be achieved since it is fabricated in specialized and controlled precast plants [1]. The use of FRC for tunnel lining can reduce the cost and labor associated with the fabrication process while it improves the post-cracking behavior considerably. The performance of steel fibers has been widely studied in the past few years, even though the corrosion factor still influences the performance of SFRC segments over its service life [2][3]. The use of polymeric fibers can counter this issue due to its non-corrosive property especially in aggressive soil environment and hydraulic or wastewater tunnels where excessive corrosion may take place.

The objective of this work is to develop the design, analysis, and verification tools for procedures that would allow for the design of precast tunnel lining using macro synthetic fibers. In addition, the effort determines the mechanical properties of fiber-reinforced sections and proposes design guidelines that allow for replacing the conventional rebars with a certain volume fraction of macro synthetic fibers. It is expected that by using appropriate mix designs and implementing material behavior in the design procedures, sustainability initiatives can be supported and documented in the context of long term performance, ductility, and cost-saving. [4, 5, 6]

Several tunnel focused research projects highlight the beneficial effect of FRC in the presence of concentrated loads and bursting [1,7]. Use of fibers increases the fatigue and impact resistance [8] of the segments in mitigation of unintentional impact loads during handling and tunnel construction operations. Segments can be manufactured using FRC only or hybrid solution combination of conventional rebar and fiber reinforcement, i.e. a hybrid system which is very attractive for larger diameter tunnels with high internal forces. A fiber only or hybrid solution depending on performance requirements may therefore present the most ideal solution. Using the current technology with high performance FRC segments, tunnel rings of more than 23 ft (7 m) diameter have been successfully used to include fibers.

Current serviceability based procedures are empirically based extensions of equivalent elastic analysis. Recent adoption of fibers as reinforcement in the Model Code 2010 [9], ACI-318 building code, and ACI 544 Fiber Reinforced Concrete Committee is promoting the use of fibers in structural concrete. Non-linear procedures such as elastic-plastic solutions close the gap between properties, analysis, modeling and service life modeling. Closed-form relationships were developed based on plastic design. The goal of Hybrid Reinforcement Concrete Research (HRC) is to reevaluate traditional reinforcing technologies by analysis of a host of available reinforcing technologies using polymeric macro fibers, and conventional reinforcements. The proposed experimental and analytical approach will lead to enhanced technology development in various areas including geotechnical engineering, transportation, water resources, pavement materials and systems.

2. RESEARCH OBJECTIVES

The intent of this comprehensive program is to determine the material properties and develop structural design guidelines for macro synthetic fibers used at different volume fractions. The test methodology will develop improved methods to characterize average residual strength in fiber-reinforced concrete mixtures made with a variety of mixtures for applications in tunneling. The goal is to test the materials in terms of optimum mix design and to develop a calibration procedure that allows for design and verification of mechanical properties. Using a system design approach where the analysis tools (e.g. structural analysis) and the design optimization tools (e.g. non-linear programming) are closely linked to design requirements[10].

3. FRC DESIGN CODES, STANDARDS AND RECOMMENDATIONS

FRC precast tunnel segments have been designed using constitutive laws recommended by international design codes and standards such as German DBV [11], RILEM TC 162-TDF [12], Italian CNR DT 204/2006 [13], Spanish EHE-08 [14], ACI-544 [15] and fib Model Code 2010 [9]. These codes propose stress-crack width or stress-strain constitutive laws of FRC as a linear post-cracking behavior (hardening or softening) or as a rigid perfectly plastic behavior based on bending test results. A majority of these design guides are related to the adoption of non-linear fracture mechanics parameters to analyze cracking phenomena and obtain allowable stress crack width relationship [16] [17]. This method is used to determine post-crack residual tensile strength from results of standard beam tests such as ASTM C1609 and EN 14651, as one of the key design parameters for FRC tunnel segments [18]. ACI 544.7R report [19] is currently the leading document for design of FRC based tunnel linings for a variety of load cases. It can be used in conjunction with ACI 544.8R report [20] to obtain the required constitutive law for the calculation of resistance moment and axial force-bending moment interaction diagrams for use with FRC precast tunnel segments [21].

4. MATERIALS CHARACTERIZATION TESTS

The large-scale testing laboratory at Arizona State University (ASU) uses the newly developed load frame that extends testing Full-scale flexure tests for applications such as tunnel lining. Figure 1 shows the schematic testing frame for large scale tunnel segments. Tests were conducted on fiber-reinforced precast tunnel segments and results were used to characterize the performance of macro polypropylene fiber-reinforced concrete (PFRC) segments with dosages of 14 lb/yd³ (8.3 kg/m³) equivalent to 0.91% volume fraction and labeled as PP14. These tests can validate the use of PFRC segments since macro-synthetic PP fibers do not suffer from corrosion that may occur in hydraulic tunnels or tunnels in presence of aggressive soil environments hence ensuring a longer service life [3]

The compressive strength was measured on cylinders in accordance with ASTM C39 cylinder testing. The tensile behavior was deduced from the flexure response through bending standard ASTM C1609 [22] which is a 4 point bend unnotched tests on 150x150x600 mm specimens (Figure 2(a)).

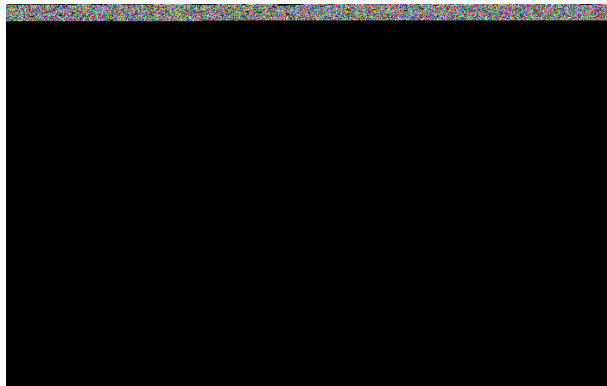


Figure 1: Load frame setup for precast tunnel segments

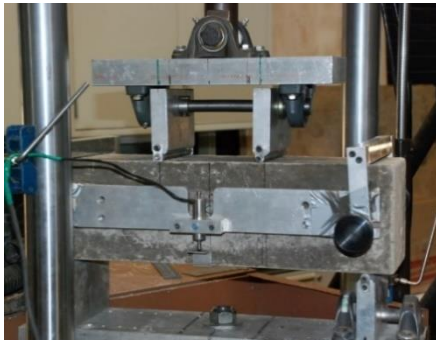
5. EXPERIMENTAL SETUP AND RESULTS

Three replicate precast tunnel segments were tested for 3-point-bending full scale flexure test, prior to which, a standard ASTM C1609 test was conducted on a total of 4 beams on the 28th day of casting tunnel lining specimens to measure the strength parameters, fiber performance, and material properties as per ACI 544.8R report and ASTM C39M test was conducted on three 4"x8" cylinders following each full scale test. The polypropylene fibers used had a nominal length of 54 mm (2.1 in) and a tensile strength of 585 MPa (85 ksi).

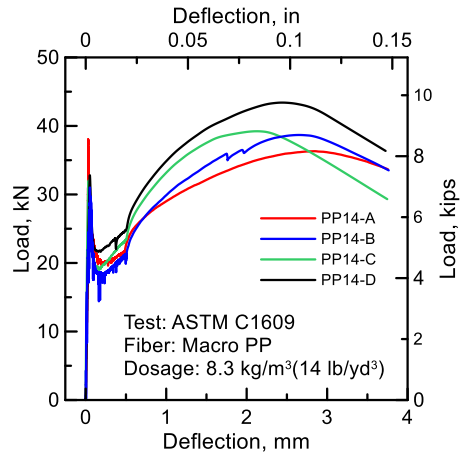
Standard ASTM C1609 beam tests were performed on four beams of size b=152 mm (6 in), h=152 mm (6 in), and span = 457 mm (18 in), to study the material properties and fiber response based on the load-deflection response of each beam. Figure 2 (b) shows the load-deflection plots of 4 PFRC beams tested on the 28th day of casting.

The full-scale three-point bending flexural tests were conducted on 3 replicate PP-FRC precast tunnel lining segments to measure the flexure performance in terms of load-deflection. Results were compared with the simulated response using the back-calculated analytical response of ASTM C1609 tests that uses the parameterized stress-strain models. The tests were conducted under load control using a loading beam and steel cylinders on top as shown in Figure 3, with a total self-weight of 220 lbs. The internal diameter of the tunnel was 2.7 m (8.75 feet) and the outer diameter was 3 m (10 feet). The dimensions of the segment were, span 'L' of 100 cm (39 in), depth 'd' of 19 cm (7.5 in) & breadth 'b' of 121 cm (48 in). The segments were instrumented with 4 LVDTs to measure the deflection, 4

strain gages to measure the strains both in compression and tension zones and 4 string sensors to measure the crack opening at the tension zone. Digital image correlation (DIC) was also set up to measure precise strains, stresses, deflection, and crack width.



(a) ASTM C1609 test setup

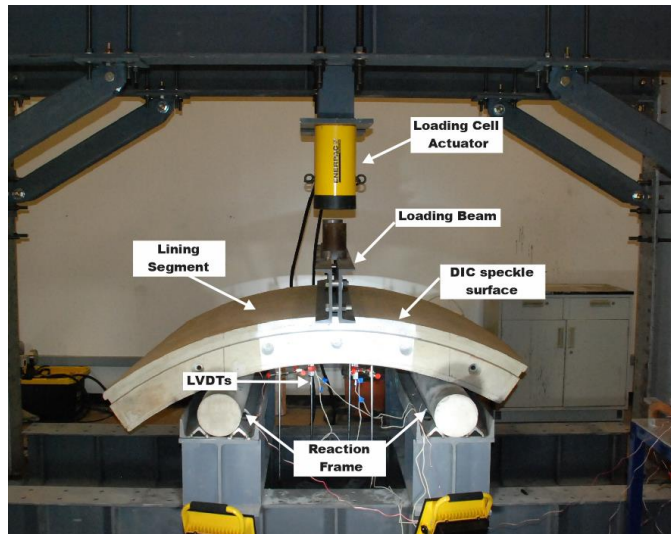


(b) PFRC with dosage 8.3 kg/m^3 test results

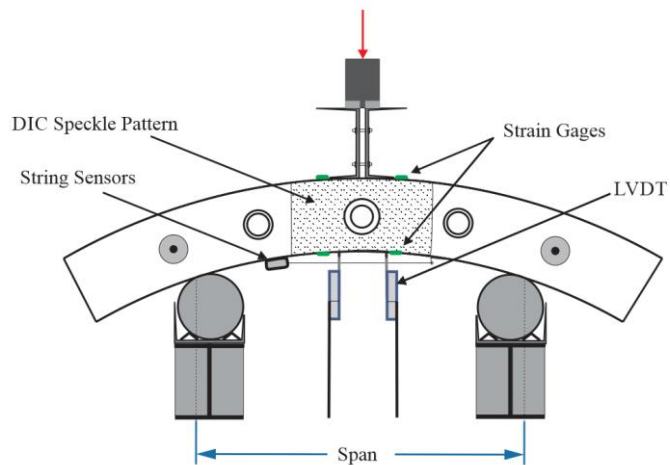
Figure 2: ASTM C1609 test results

6. TEST RESULTS OF PFRC SEGMENTS

A plot comparing the load vs. deflection response of the three replicate PFRC segments is shown in Figure 4. PP14_N4 shows the highest first crack strength and post crack strength. PP14_N1 and PP14_N4 show strain hardening behavior with higher post-peak response compared to the first crack strength. All the specimens reach maximum post-peak strength at around 4 mm deflection. The large-scale test results also correlate with the C1609 tests showing a similar response with a strain-hardening behavior. Figure 5 (a) and (d) show the failure of the tunnel segments with the crack opening and fibers straining in the tension zone. Table 1 summarizes the test results of all the tested PP-FRC segments.



(a) Flexure Test Setup



(b) Schematic diagram of the test setup with the Instrumentations

Figure 3. Large scale load frame and instrumentations for flexure testing

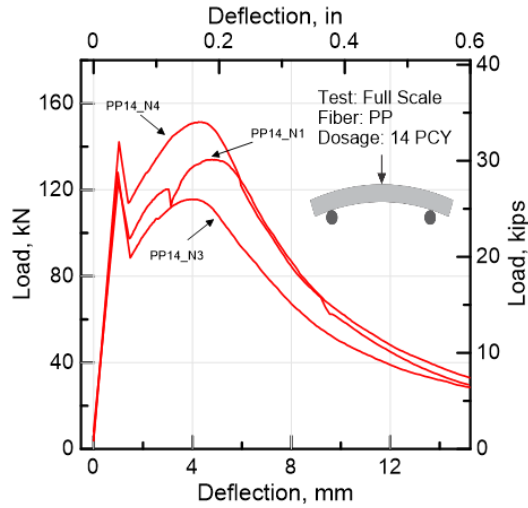
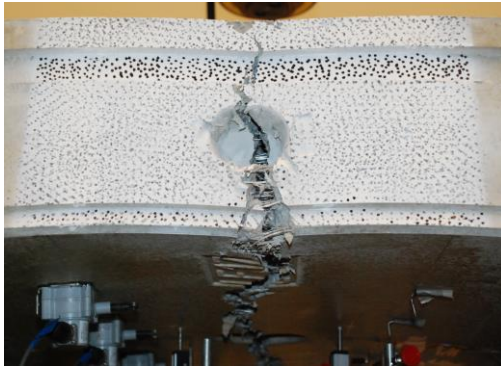


Figure 4. Load vs. deflection comparison of all three PFRC segments



a) Cracked PFRC specimen



b) Intrados of a cracked specimen

Figure 5. Flexure test failure details

Fiber Type	SampleID	First Crack Load, P_{cr} , (kN)	First Crack Deflection, δ_{cr} , (mm)	Load at L/300 deflection (3.3 mm) (kN)	Load at L/150 deflection (6.6 mm) (kN)	Maximum Load, P_{max} (kN)	Deflection at Max. Load, δ_{max} (mm)	P_{max}/P_{cr}
PP14	PP14_N1	124	1.00	119	111	134	4.83	1.08
	PP14_N3	128	0.99	113	85	128	0.99	1
	PP14_N4	142	1.02	145	108	151	4.32	1.06
	Average	131	1.00	126	102	138	3.38	1.05
	σ_{std}	10	0.01	17	14	12	2.1	0.04
	CV(%)	7	1	14	14	9	62	4

Table 2: Experimental test results of all 3 tunnel lining segments

Figure 6 shows plots of the data collected from the instrumentations from one of the tests. The DIC method uses a non-contact means of measuring displacements results in precise mid-span deflections compared to LVDTs mounted within 10 cm from the centerline. Figure 6 (d) shows the agreement between the deflection measured from LVDTs and DIC. Figure 6 (c) shows the strain vs. time plot in which two strain gages in the compression and two strain gages in tension results are presented. A common trend is also observed in all other panels with the compressive strain always higher than the tensile strain, this can be because the strain gages in the tension zone were attached near the grout hole resulting in lower measure of strain.

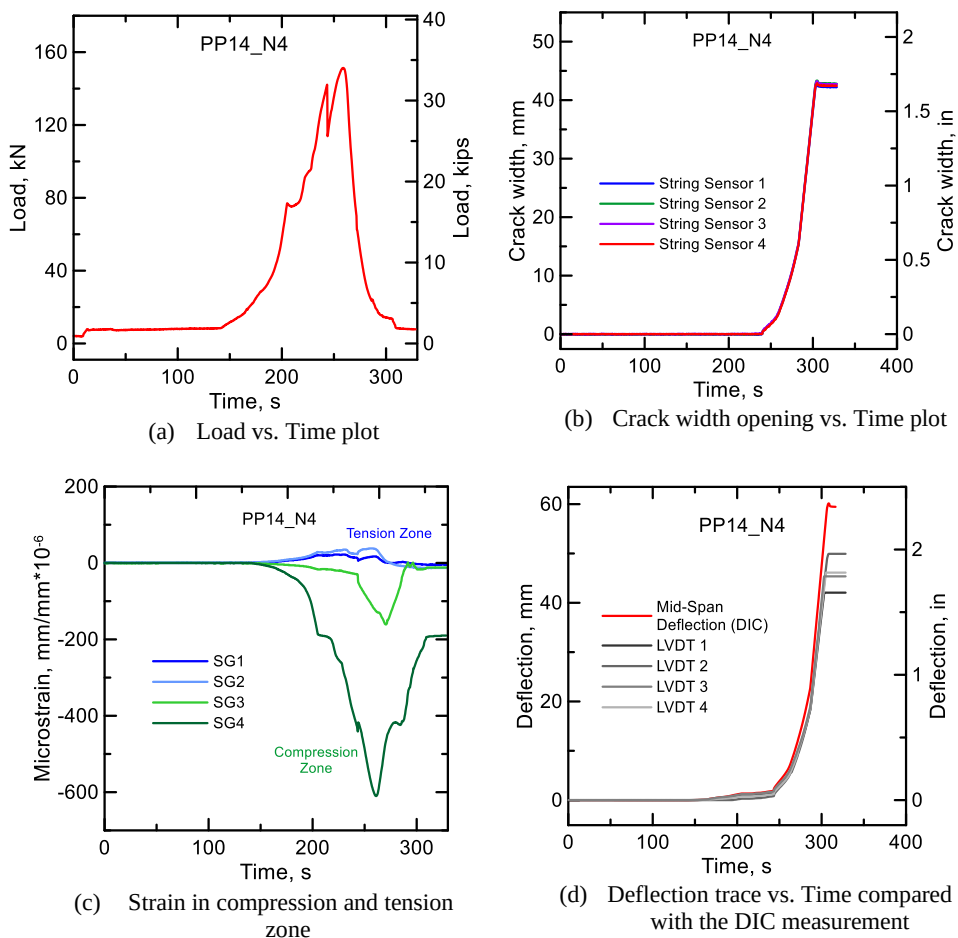


Figure 6. Time history plots of PP14_N4 for various instrumentations

7. MODEL SIMULATION AND CORRELATIONS

The ASTM C1609 test results can be used to extract the material properties by back-calculating the load-deflection plot and using closed-form solutions as proposed in ACI 544.8R report [10]. A quad-linear tension model can be used for precise back-calculation of the C1609 results based on the complete load-deflection response. The Tensile stress-strain and compression stress-strain models used for back-calculation are shown in figure 7. The tensile curve has four linear segments for strain softening/hardening and the compression response has two linear terms for an elastic perfectly plastic model. Two of the ASTM C1609 beam test results were back-calculated using the closed-form equations based on these stress-strain models and simulations were generated as shown in figure 8 (d). The stress-strain models used to back-calculate the curves are shown in figure 8 (a-b). The elastic range however shows a higher stiffness compared to the experimental data. This is because the deflections of the panels were measured with respect to the strong floor and not the centerline of the specimen at the support. The observed discrepancies are therefore attributed to the spurious deformations and the settlement of the rollers at the sliding bearings that support the curved-shaped specimen along a thin contact line. While this initial compliance of the entire support frame and the specimen can be adjusted to account for the LVDT that was placed on the ground, the observed differences serve to show that the initial stiffness values are not a direct representation of the elastic modulus of the specimen.

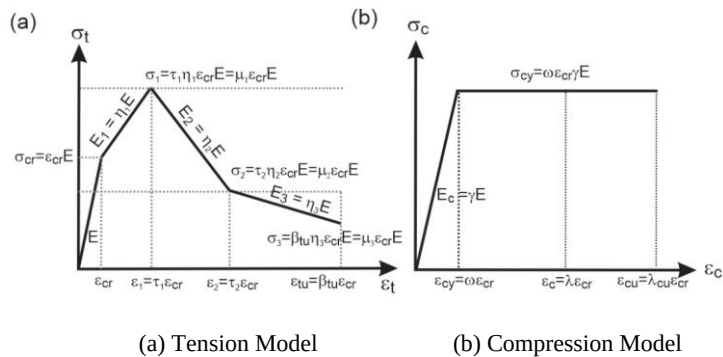
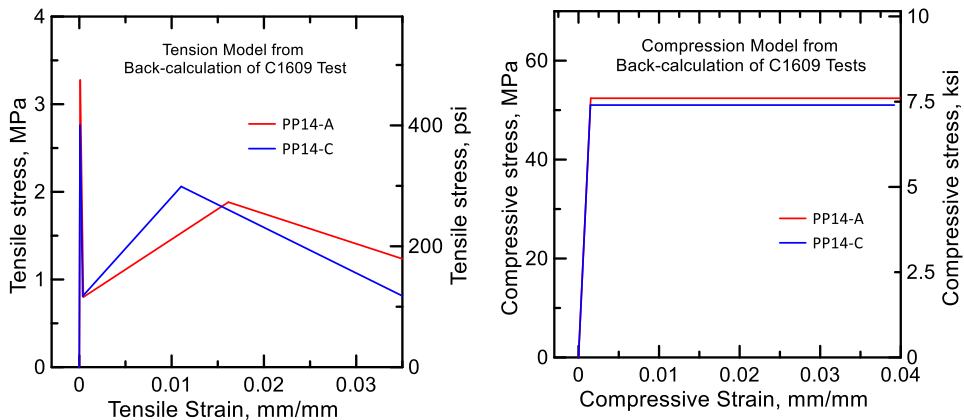


Figure 7. stress-strain material models



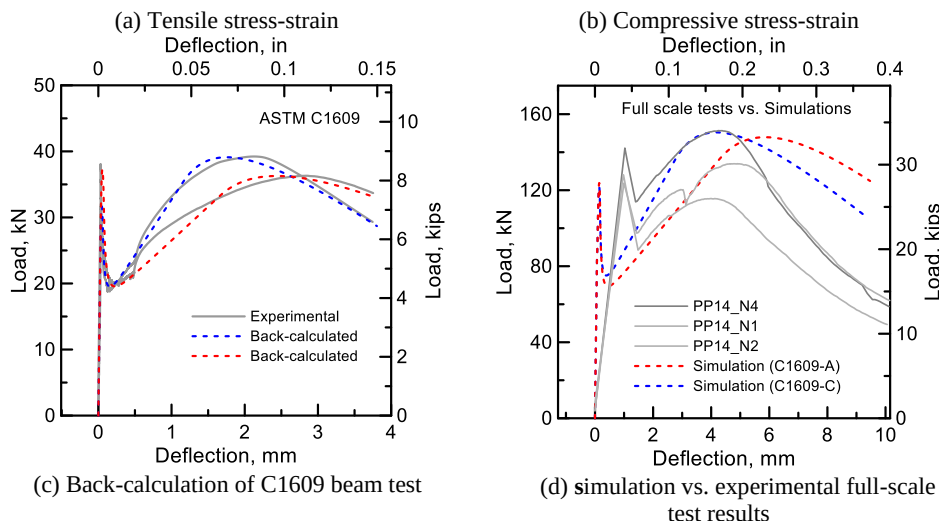


Figure 8. Material properties extracted from ASTM C1609 tests of PFRC beams

8. CONCLUSIONS

Polymeric fibers can be used as alternative reinforcement of tunnel segments as validated by sufficient post-crack strength when only PFRC segments were tested. The study conducted was also able to validate the method used for predicting full-scale responses that use ASTM C1609 tests to back-calculate and extract material parameters which then uses closed-form solutions to predict the full-scale response of the flexure tests. This method of testing and verification using the standard flexural tests can further be used in designing sections with any depth, fiber dosage or hybrid sections.

REFERENCES

- [1] de Waal, R.G.A., 1999, "Steel Fiber Reinforced Tunnel Segments", *Ph.D Dissertation*, Delft University of Technology, The Netherlands.
- [2] Nehdi, M., Abbas, S., Soliman, A. (2015). Exploratory study of ultra-high performance fiber reinforced concrete tunnel lining segments with varying steel fiber lengths and dosages. *Engineering Structures*. 101. 733-742. 10.1016/j.engstruct.2015.07.012.
- [3] Conforti, A., Trabucchi, I., Tiberti, G., Plizzari, G., Caratelli, A. and Meda, A. Precast tunnel segments for metro tunnel lining: A hybrid reinforcement solution using macro-synthetic fibers. *Engineering Structures*, Volume 199, 15 November 2019,
- [4] Plizzari G.A.; Tiberti G., 2007, "Structural Behaviour of SFRC Tunnel Segments", *Proceedings of the 6th International Conference on Fracture Mechanics of Concrete and Concrete Structures (FraMCoS 2007)*, A. Carpinteri, P. Gambarova, G. Ferro, and G.A. Plizzari, eds. Catania, Italy, V. 3, pp. 1577-1584.
- [5] Plizzari G.A.; and Tiberti G., 2006, "Steel Fibers as Reinforcement for Precast Tunnel Segments", *Tunnelling and Underground Space Technology*, V. 21, No. 3-4, pp. 438-439.

- [6] De la Fuente, A.; Pujadas, P.; Blanco, A.; and Aguado, A., 2012, “Experiences in Barcelona with the Use of Fibres in Segmental Linings”, *Tunnelling and Underground Space Technology*, V. 27, No. 1, pp. 60-71.
- [7] Schnütgen, B., 2003, “Design of Precast Steel Fiber Reinforced Tunnel Elements, *Proceedings of the RILEM TC 162-TDF Workshop, Test and Design Methods for Steel Fiber Reinforced Concrete – Background and Experiences*, Bochum, Germany, pp.145-152.
- [8] di Prisco, M.; and Felicetti, R., 2004, “On Fatigue of Plain and Fibre-Reinforced Concrete Ground Slabs”, *6th Int. RILEM Symposium, (BEFIB'2004)*, Varenna, Italy, pp. 1195-1206.
- [9] Fédération internationale du béton. 2013. Fib model code for concrete structures 2010. <http://site.ebrary.com/id/10780745>.
- [10] ACI 544.8R-16: Report on Indirect Method to Obtain Stress-Strain Response of Fiber-Reinforced Concrete (FRC), ACI Committee 544 ACI 544.8R (2016)
- [11] DBV, 1992, “Design Principles of Steel Fibre Reinforced Concrete for Tunnelling Works”, *DBV Recommendation*, German Concrete Association, pp. 19–29.
- [12] RILEM TC 162-TDF, 2003, “Test and Design Methods for Steel Fibre Reinforced Concrete. □–□ Design Method: Final Recommendation”, *Materials & Structures*, 36, 262, pp. 560–567.
- [13] CNR DT 204/2006, 2006, “Guidelines for the Design, Construction and Production Control of Fibre Reinforced Concrete Structures”, Italian National Research Council – CNR.
- [14] EHE-08, 2008, “Code on Structural Concrete, ANNEX 14 – Recommendations for Using Concrete with Fibres”, Ministry of Public Works and Transport, Madrid, Spain, pp. 1-55.
- [15] ACI 544.7R-16 Report on Design and Construction of Fiber-Reinforced Precast Concrete Tunnel Segments ACI Committee 544 ACI 544.7R (2016)
- [16] Hillerborg, A.; Modker, M.; and Petersson P.E., 1976, “Analysis of Crack Formation and Crack Growth in Concrete by Means of Fracture Mechanics and Finite Elements”, *Cement and Concrete Research*, V. 6, pp. 773–782.
- [17] Soranakom, C., and Mobasher, B., “Flexural Design of Fiber Reinforced Concrete,” *ACI Materials Journal*, September-October 2009, pp. 461-469
- [18] ACI 544.6R-15 Report on Design and Construction of Steel Fiber-Reinforced Concrete Elevated Slabs, Committee Chair, ACI Committee 544 ACI 544.6R (2015)
- [19] ACI 566.7R-16 Report on Design and Construction of Fiber-Reinforced Precast Concrete Tunnel Segments ACI Committee 544 ACI 544.7R (2016)
- [20] 544.9R-17: Report on Measuring Mechanical Properties of Hardened Fiber-Reinforced Concrete, ACI Committee 544 ACI 544.8R (2017)
- [21] Yao, Y., Mobasher, B., Soranakom, C. “Closed Form Solutions for Flexural Design of Hybrid Steel Fiber Reinforced Concrete Beam”, *Engineering Structures*; 100:164-177 · September 2015.
- [22] ASTM Standard C1609/C 1609M, “Standard Test Method for Flexural Performance of Fiber-Reinforced Concrete (Using Beam With Third-Point Loading)”, ASTM International, West Conshohocken, PA, 2007.